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YASAK FUR FRONTIERS, YAKUT HERDERS AND THE AMUR-MARITIME ANNEXATIONS

Minusinsk Kurgans, Bohai-Jurchen Towns, 17th-Century Yasak Frontiers and Russian Settlement in Eastern and Pacific Siberia

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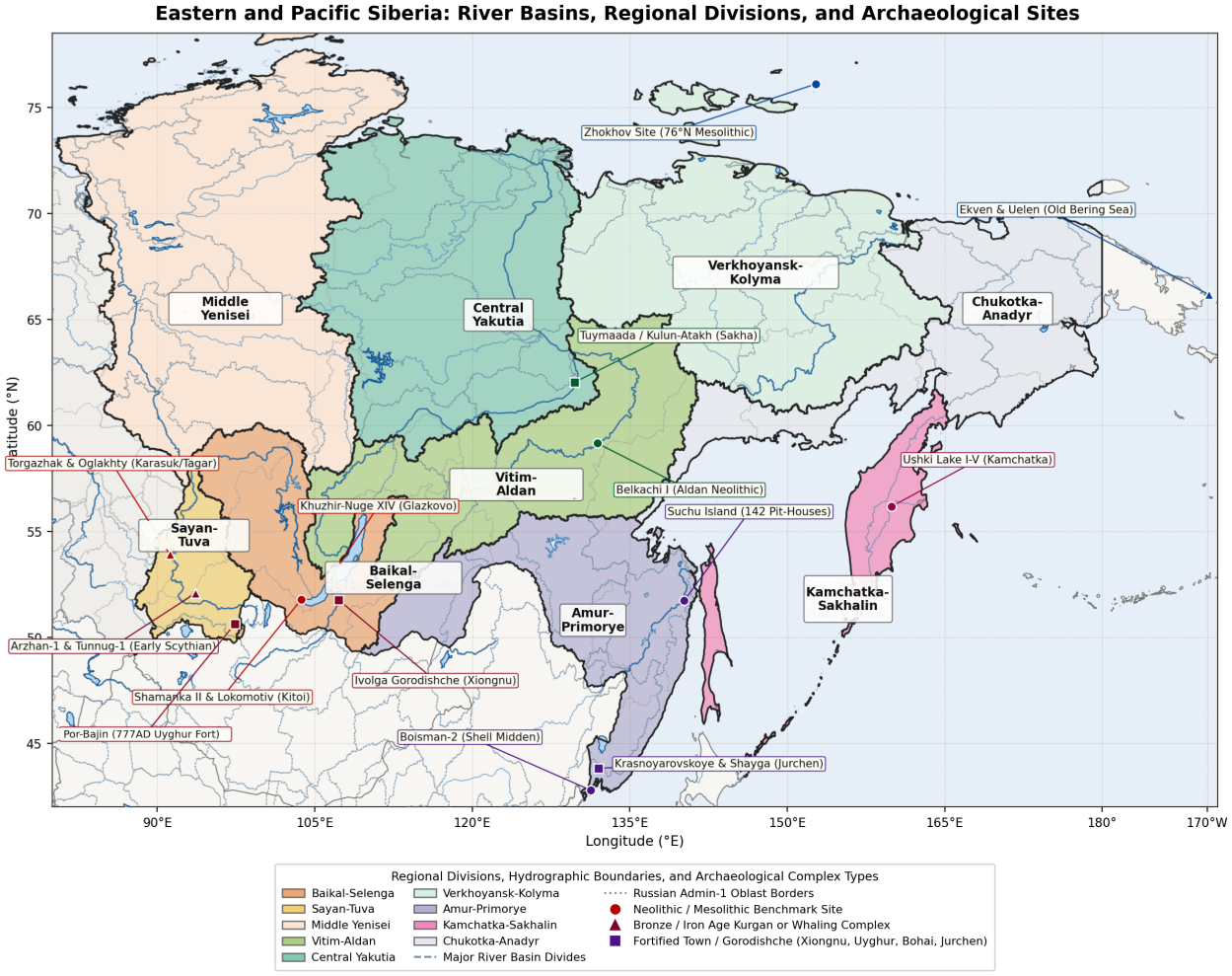


Figure 1. The nine drainage regions of Eastern and Pacific Siberia (10,11M km^2), grouped into the Yenisei-Baikal (3,28M km^2), Lena-Kolyma (4,05M km^2) and Pacific (2,78M km^2) basins.

Abstract

Eastern and Pacific Siberia cover ~10,11M km^2 across three major river and coastal systems and nine regions (Figure 1): the Yenisei and Baikal Region (Baikal-Selenga, Sayan-Tuva and Middle and Lower Yenisei), the Lena and Kolyma Region (Vitim-Aldan, Central Yakutia and Middle Lena and Yana-Indigirka-Kolyma) and the Pacific Region (Amur-Primorye, Kamchatka-Sakhalin-Kurils and Chukotka-Anadyr) [1]. For several millennia after the Ice Age, fishing and hunting communities along the

Baikal-Angara rapids, the Lower Amur and the Pacific coast grew slowly from ~15,2k in 10000BC to ~56,7k by 4500BC [2][3][4]. From the Bronze Age onwards, millet farming and stockbreeding in the Minusinsk Hollow, Tuva, the Selenga steppes and Primorye raised the combined population to ~201,3k during the Tagar, Slab Grave and Yankovsky Iron Age (500BC) and ~265,9k during the medieval Bohai, Uyghur and Yenisei Kyrgyz period (800AD) [5][6][7][8]. When Russian Cossacks crossed the Yenisei, Lena and Amur in the 17th century, fur-tribute (*yasak*) books recorded 220,53k native inhabitants in 1650AD (234,8k total population by 1700AD), including 28,47k Sakha cattle-and-horse herders in the Central Yakutian *alaas* lowland and 27,3k Buryat pastoralists around Lake Baikal [9][10]. Over the 18th and 19th centuries, natural growth among the Buryat and Sakha peoples together with Russian peasant, Cossack and gold-mining settlement across the Minusinsk, Irkutsk, Transbaikal and post-1858 Amur-Ussuri basins lifted the population to 390,8k in 1750AD, 669k in 1800AD, 1,17M in 1850AD and 2,45M by 1900AD [11][12][13].

Background

Eastern and Pacific Siberia span a wide range of environments, from the sheltered steppes of the Minusinsk Hollow, Tuva and the Selenga basin and the monsoon forests of the Amur, Ussuri and Primorye plains to the lake-meadow depressions (*alaas*) of Central Yakutia, the permafrost larch taiga of the Tunguska, Vitim, Aldan and Kolyma basins and the salmon rivers and sea-mammal coasts of Kamchatka, Sakhalin, the Kurils and the Bering Strait [1]. Before the 17th century, crop growing was limited to the southern steppe valleys and the Middle Amur, Ussuri and Primorye lowlands. Across the remaining 8,5M km^2 of northern forest and Arctic tundra, population density depended on freshwater and salmon runs, coastal sea-mammal hunting, wild deer hunting and - from the Middle Ages onward - horse, cattle and reindeer herding.

Four bodies of historical and archaeological research underpin the estimates presented here. In the Lake Baikal basin, the Baikal Hunter-Gatherer Project (BHGP) obtained over 538 accelerator mass spectrometry (AMS) radiocarbon dates paired with $\delta^{13}C$ and $\delta^{15}N$ isotope measurements from burials at Shamanka II, Lokomotiv, Ust'-Ida I, Kurma XI and Khuzhir-Nuge XIV. Correcting for old carbon offsets of 300-700 ^{14}C years in fish and seal consumers confirmed a 700-year gap in formal cemetery burials (4900-4200 cal BC) between the Early Neolithic Kitoi culture and the Late Neolithic Isakovo-Serovo culture [2][14][15][16]. In the Minusinsk Hollow and Tuva, AMS dating and isotope studies of 354 individuals show that broomcorn millet (*Panicum miliaceum*) became a staple crop during the Karasuk (~1400-900 cal BC) and Tagar (~800-100 cal BC) periods, supporting a three- to fourfold rise in population reflected in tens of thousands of stone-walled burial mounds (*kurgans*) and permanent winter villages [5][17][18][19][20][21].

In the Amur-Primorye basin and the Northwest Pacific islands and coasts, excavations of early pottery sites (Osipovka, Gromatukha), large Neolithic pit-house villages (Suchu Island with 142 house depressions; Boisman-2 shell midden), Bohai and Jurchen walled towns (Krasnoyarovskoye covering 180 ha; Shayginskoye with over 310 excavated heated houses), 380 Kuril Island radiocarbon dates and Bering Strait whalebone villages (Ekven and Uelen with over 530 graves) provide direct house and burial counts [3][8][22][23][24][25][26][27][28][29]. B. O. Dolgikh's clan-by-clan study of 17th-century *yasak* tax books [9], supplemented by regional studies of Khongorai, Yakutsk *uyezd*, Priamurye and Kamchatka-Chukotka [10][30][31][32][33][34], establishes the mid-17th-century native population. For the 18th and 19th centuries, the ten Poll-Tax Revisions (*revizii* of 1719-1858), the records of the 1763 *Yasak* Commission and post-1822 Steppe Dumas, provincial statistical reviews and the First All-Russian Census of 28 January 1897 document both native population growth and the Russian farming and urban settlement of Eastern Siberia and the Far East [11][12][13][35][36][37][38].

Methodology

Regional Divisions

All population estimates are organised across the nine drainage regions defined by Chernykh et al. [1] and mapped in Figure 1. The Yenisei and Baikal Region (~3,28M km^2) comprises the Baikal and Selenga Basin (~650k km^2: Lake Baikal, Upper Angara, Barguzin, Selenga, Uda, Khilok, Chikoy and Onon-Ingoda headwaters), the Sayan, Minusinsk and Tuva Basin (~380k km^2: Minusinsk Hollow, Abakan, Chulym, Kan and Upper Yenisei/Ulug-Khem/Khemchik basins in Tuva) and the Middle and Lower Yenisei (~2,25M km^2: Middle and Lower Yenisei, Lower Angara, Stony and Lower Tunguska, Pyasina, Khatanga and Taimyr Peninsula).

The Lena and Kolyma Region (~4,05M km^2) covers the Vitim and Aldan Basins (~1,12M km^2: Vitim, Olekma, Aldan, Amga, Uchur and Maya basins), Central Yakutia and the Middle Lena (~1,38M km^2: Middle and Lower Lena, Central Yakutian *alaas* lowland and Vilyuy basin) and the Yana, Indigirka and Kolyma Basins (~1,55M km^2: Yana, Indigirka, Alazeya, Kolyma and Omolon basins, plus the New Siberian Islands). The Pacific Region (~2,78M km^2) covers the Amur, Ussuri and Primorye Basin (~980k km^2: Shilka-Argun, Upper/Middle/Lower Amur, Zeya, Bureya, Ussuri, Lake Khanka and Primorye coast), Kamchatka, Sakhalin and the Kuril Islands (~650k km^2: Okhotsk coast, Kamchatka Peninsula, Sakhalin Island and Kuril Archipelago) and Chukotka, Anadyr and the Bering Strait (~1,15M km^2: Anadyr basin, Chukotka Peninsula, Bering Strait coast and Koryak Highlands/Penzhina-Gizhiga bays).

Estimation Methods

Four standard archaeological and historical methods are used to convert field and tax records into regional population totals across nineteen benchmark dates from 10000BC to 1900AD. Using floor-area ratios (~6-10 m^2 per person) and North Asian accounts [29][39], single-hearth pit-houses of 20-45 m^2 (Boisman, Yankovsky and Shayga *kan*-heated houses) are estimated at 5-7 occupants; medium multi-hearth houses of 60-120 m^2 (Suchu Island, Okhotsk Culture and Ekven whalebone *nynglu*) at 12-20 occupants; and large communal winter houses (Karasuk timber houses at Torgazhak of 100-200 m^2; Itelmen winter *zimniki* of 150-300 m^2) at 25-80 occupants. For walled towns (*gorodishcha*) of the Xiongnu (Ivolga, Bayangol), Uyghur Khaganate (Por-Bajin, Shagonar), Bohai Kingdom (Kraskinskoye, Koksharovka-1) and Jurchen Jin/Eastern Xia (Krasnoyarovskoye, Shayginskoye), excavated house counts (54 houses in 25% of the 7,2 ha interior at Ivolga; over 310 houses across 47,6 ha at Shayginskoye) indicate 30-55 houses/ha in residential quarters, corresponding to 150-300 persons/ha in closely built towns and 40-80 persons/ha in large mountain refuge enclosures [6][8][26].

Living community size P is estimated from the minimum number of individuals (M_{MNI}) buried over a calibrated cemetery duration T (in years) at an annual mortality rate m of 0,03-0,035 per year via $P = (M_{\mathrm{MNI}} \cdot c_{\mathrm{exc}})/(m \cdot T)$, where c_{exc} accounts for unexcavated cemetery area and child burial under-representation [15]. In the historical period, adult male tribute-payers (*yasachnye lyudi*, aged 18-50) in 17th-century Russian *yasak* registers are converted to total population using B. O. Dolgikh's household multipliers [9]: x 3,5-4 for forest and tundra hunter-herders (Evenks, Evens, Yukaghirs, Kets, Samoyeds), x 4-4,5 for pastoralists (Sakha, Buryats, Yenisei Kyrgyz, Tuvans) and x 4,5-5 for settled plough farmers and multi-family river fishers (Daurs, Duchers, Itelmens). For 1750, 1800, 1850 and 1900AD, district (*uyezd* / *okrug*) returns from the 2nd (1744), 5th (1795) and 9th-10th (1850-1858) revisions, Steppe Duma registers and the 1897 Census (adjusted to 1900AD using provincial yearbooks) are allocated to the nine drainage regions [11][12][13][40].

Results

Baikal and Selenga Basin

The Early Neolithic Kitoi Culture and the Middle Holocene Gap

Following late Ice Age and Mesolithic hunting and fishing along the Angara rapids and Selenga river terraces (Ust'-Belaya, Studenoe-1/2, ~8000-5800 cal BC; regional population rising from ~2,2k in 10000BC to ~3,8k in 8000BC), the Early Neolithic Kitoi culture (*Kitoyskaya kultura*, reservoir-corrected to ~5750-4950 cal BC) established large formal cemeteries at productive fishing and sealing spots [2][14][15]. At Shamanka II on the Kultuk peninsula at the south-western tip of Lake Baikal, excavations uncovered 112 graves containing 155 individuals (116 AMS dates), with over 85% of burials belonging to the Kitoi phase (~5650-4950 cal BC) and a smaller Early Bronze Age Glazkovo group (13 individuals, ~2600-1950 cal BC) [15]. At Lokomotiv at the Irkut-Angara confluence in Irkutsk, excavations of 91 graves yielded 149 individuals (~5750-5000 cal BC), whose high δ^{15}N values (+13,5‰ to +16,8‰) reflect heavy reliance on Angara grayling, sturgeon and Baikal seal (*Pusa sibirica*) [2][14].

Across all 538 reservoir-corrected AMS dates from Cis-Baikal cemeteries (Figure 2, Panel A), no formal cemetery burials date between 4900 and 4200 cal BC [2][16]. Large Kitoi fishing communities broke up into smaller mobile hunting bands (regional population falling from ~8,2k in 6000BC to ~3,2k in 4500BC), before genetically distinct Late Neolithic Isakovo-Serovo (~4200-3000 cal BC; Ust'-Ida I with 67 individuals) and Early Bronze Age Glazkovo (~3000-1950 cal BC; Khuzhir-Nuge XIV with 79 graves and 89 individuals, Kurma XI and Sagan-Zaba II sealing camps) communities raised the regional population to ~7,5k in 3000BC and ~12k in 2000BC [2][41].

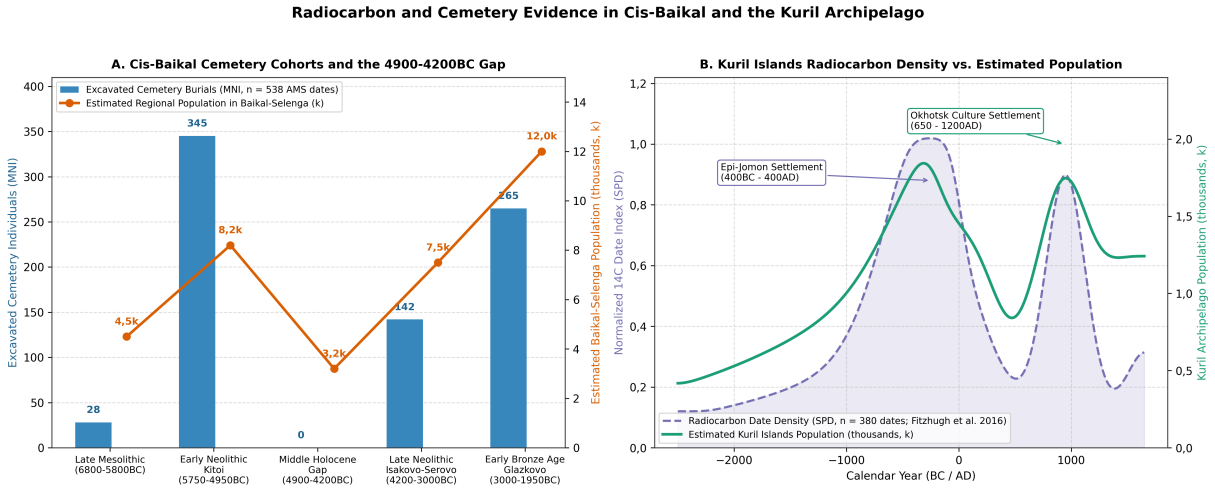


Figure 2. Panel A plots minimum numbers of individuals (MNI) and reservoir-corrected AMS dates across Cis-Baikal cemeteries (Shamanka II, Lokomotiv, Ust'-Ida I and Khuzhir-Nuge XIV), highlighting the 4900-4200 cal BC burial hiatus between the Kitoi and Isakovo-Serovo cultures. Panel B shows summed radiocarbon date counts across the Kuril Islands (380 dates), marking the Epi-Jomon (400BC - 400AD) and Okhotsk Culture (650 - 1200AD) coastal settlement waves.

The Slab Grave Culture, Khirigsuurs and the Xiongnu Frontier

The spread of horse, sheep, goat and cattle herding during the Late Bronze and Early Iron Ages reshaped settlement across the Selenga, Uda, Khilok, Dzhida and Onon-Ingoda steppes. More than 3k stone-slab burial enclosures of the Slab Grave culture (*Kultura plitochnykh mogil*, ~1300-300BC) and large *khirigsuur* stone mounds are recorded across Buryatia and western Transbaikalia, supporting an estimated regional population of ~18,5k by 1200BC and ~27k by 500BC.

From the late 2nd century BC to the 1st century AD, the Xiongnu Empire (*Khunnu*) held the Selenga-Dzhida basin as a northern farming, ironworking and garrison frontier [6][42]. At Ivolga Fortress (*Ivolginskoye gorodishche*, 14 km south-west of Ulan-Ude), enclosed by four concentric ramparts and ditches (348 x 216 m, ~7,2 ha preserved), A. V. Davydova's excavations of roughly a quarter of the interior exposed 54 semi-subterranean houses heated by L-shaped stone and clay flues (*kan*), iron-smelting furnaces, grain storage pits containing millet, barley and wheat and the adjacent 216-grave Ivolga cemetery; scaling across the unexcavated quarters indicates ~300-380 contemporary houses, or a garrison and craft town of ~2,5k-3k residents [6]. Beyond Bayangol (~10 ha), Dyrestui and Ilmovaya Pad (over 320 Xiongnu graves including elite terraced tombs), open farming and craft villages such as Nizhnemangirtuyskoe (1,8 ha) and Dureny I-II (stretching along 12 km^2 of the Chikoy valley) supplied grain and iron tools to nomadic camps, supporting a regional population of ~36k around 1AD [42].

Medieval Kurykans, Buryat Steppe Dumas and Russian Settlement

Between the 6th and 11th centuries AD, the Kurykan (*Kurumchinskaya*) confederation (the *Gulifan* of Tang Chinese chronicles, recorded as fielding up to 5k horsemen) built more than 40 stone-walled hillforts (*shibe* on Olkhon Island and in the Barguzin, Kuda and Angara valleys), bloomery iron furnaces and hay-meadow irrigation ditches, sustaining ~34k-35,5k people across the Baikal and Selenga basin between 800 and 1200AD. In the mid-17th century, B. O. Dolgikh [9] and V. N. Davydov et al. [43] reconstructed 27,3k Buryats (~13,5k western Bulagat, Ekhirit and Khongodor Buryats along the Angara, Oka and Olkhon, and ~13,8k eastern Khori, Selenga and Tabunut Buryats) together with 9,3k Baikal and Daurian Evenks, giving a native population of 36,6k in 1650AD (0,0563 persons/km^2), rising to 49k by 1700AD after the founding of Irkutsk (1661), Selenginsk (1665) and Verkhneudinsk (1666).

During the 18th and 19th centuries, the Baikal-Selenga basin became the most populous part of Eastern Siberia [35][44]. Following the 1727 Treaty of Kyakhta (which fixed the Russo-Qing border and organised six Buryat border-guard regiments) and M. M. Speransky's 1822 *Statute on Indigenous Administration* (*Ustav ob upravlenii inorodtsev*, which created the Kudinskaya, Balaganskaya, Idinskaya, Verkholenskaya, Olkhonskaya, Tunkinskaya and Alarskaya Steppe Dumas in Irkutsk Governorate and the Khorinskaya, Selenginskaya, Barguzinskaya and Aginskaya Steppe Dumas in Transbaikalia), Buryat households increasingly combined livestock herding with haymaking and rye and wheat growing [35]. Returns compiled by P. I. Keppen, Yu. A. Gagemeister and S. K. Patkanov show the Buryat population growing from ~27,3k in 1650 to ~73k at the 3rd Revision (1763), ~115k in 1800, ~195k at the 9th Revision (1850) and 288,6k in the 1897 Census (108,9k in Irkutsk Governorate and 179,7k in Transbaikalia Oblast) [12][44].

From the 1760s, tens of thousands of exiled Old Believer peasants (*Semeiskie*) settled in the fertile Chikoy, Khilok and Uda valleys of western Transbaikalia, while Russian peasant villages multiplied along the Moscow Highway (*Moskovskiy trakt*) toward Irkutsk (which grew to 51,5k inhabitants in 1897), Verkhneudinsk, Chita and the tea-trade town of Kyakhta-Troitskosavsk [11][40]. Combined total population in the Baikal and Selenga basin rose from 105k in 1750AD to 210k in 1800AD, 415k in 1850AD and 850k by 1900AD (~430k in the Upper Angara, Irkutsk, Balagansk and Verkholensk districts of Cis-Baikal and ~420k in the Selenga, Barguzin, Verkhneudinsk, Troitskosavsk, Chita and Aksha districts of western Transbaikalia).

Sayan, Minusinsk and Tuva

Afanasievo, Okunev, Andronovo and Karasuk Herders and Farmers

After Upper Palaeolithic and Neolithic hunting and fishing along the Yenisei terraces (Afontova Gora, Kokorevo, Toora-Dash; regional population growing from ~2,5k in 10000BC to ~8,2k in 4500BC), the sheltered steppes of the Minusinsk Hollow and Tuva (~380k km^2) supported dense Bronze Age herding and farming populations [5][19]. Across the Afanasievo (~3100-2500 cal BC), Okunev (~2500-1800 cal BC) and Andronovo/Fedorovo (~1800-1400 cal BC) periods, radiocarbon modelling of 88 new and over 300 previously published human-bone ^{14}C dates [19] tracks the shift from early C$_3$-diet herders ($\delta^{13}\mathrm{C} = -18{,}9 \pm 0{,}6‰$) to bronze-working pastoral communities (~14k in 3000BC and ~22k in 2000BC). During the Karasuk culture (*Karasukskaya kultura*, ~1400-900 cal BC), human bone carbon isotope values shift sharply to $\delta^{13}\mathrm{C} = -14{,}8 \pm 2{,}1‰$, showing widespread cultivation of C$_4$ broomcorn millet alongside sheep and cattle herding [5]. More than 10k stone-slab burial enclosures (over 2,5k excavated across more than 100 cemeteries; Sukhaya Erba alone contains over 600 enclosures) and permanent winter villages such as Torgazhak on the Askiz River (6 excavated timber-and-sod pit-houses of 100-200 m^2 each, housing ~120-180 villagers) indicate a Late Bronze Age population of ~36k around 1200BC [17].

Tagar, Aldy-Bel and Uyuk-Sagly Kurgans and Tashtyk Crypts

During the Tagar culture (*Tagarskaya kultura*, ~800-100 cal BC), the Minusinsk basin held the largest concentration of Early Iron Age burial mounds in Siberia, with 25k-30k recorded kurgans across the Podgornovo, Saragash and Tes stages, including the royal mound of Salbyk (70 x 70 m stone enclosure with slabs weighing 30-50 tonnes) and clan-marked enclosures below Mount Tepsei [18][45]; collective timber burial vaults of the Saragash and Tes phases hold 30-150+ individuals per kurgan. In Tuva during the Aldy-Bel and Uyuk-Sagly cultures, satellite and geomagnetic surveys in the Uyuk River valley ("Siberian Valley of the Kings") have mapped thousands of stone and timber kurgans [20][21]. At Tunnug-1 (Arzhan-0, 9th century BC) and Arzhan-1 (~800 cal BC, 120 m diameter, 70 radial larch chambers containing over 160 sacrificed riding horses) [46], construction required an estimated 1k-1,5k workers, and combined Minusinsk (~36k) and Tuva (~22k) populations reached ~58k around 500BC. In the subsequent Tashtyk (*Tashtykskaya kultura*, 1st-7th centuries AD) and Kokel periods, known for the permafrost burials at Oglakhty (painted plaster funerary masks and leather effigies) and collective burial crypts (*sklepy*) containing 20-120+ individuals [47], the region sustained ~55k in 1AD and ~52k in 500AD.

Yenisei Kyrgyz, Uyghur Fortresses, Khongorai and Imperial Settlement

Along the Khemchik and Ulug-Khem valleys of Tuva between 745 and 840AD, the Uyghur Khaganate built 17 rectangular stamped-clay (*pakhs*) fortresses linked by earthen walls (*Dlinnaya Stena*), including Shagonar I-V (Shagonar III covers 22 ha), Bazhyn-Alaak (18 ha) and the island complex of Por-Bajin (215 x 162 m, 3,5 ha in Lake Tere-Khol) [7][48]. Using the 774-775AD radiocarbon spike, Kuitems et al. [49] dated the felling of Por-Bajin's foundation larch timbers to the summer of 777AD under Tengri Bögü Khagan. North of the Sayan Range from the 6th to 13th centuries AD, the Yenisei Kyrgyz state - supported by plough farming, irrigation canals (*aryks*) along the Abakan, Uybat and Erba rivers and elite *chaatas* cemeteries (Uybatsky, Kopensky) - overthrew the Uyghur Khaganate in 840AD [50]. Around 800-900AD, the Sayan-Tuva region supported ~64k inhabitants (~40k in Minusinsk and ~24k in Tuva), before 13th-century Mongol campaigns (1207-1293AD) and deportations reduced the population to ~42k in 1200AD and ~28k in 1500AD.

In 1650AD, the four Yenisei Kyrgyz princely *uluses* (Altysar, Altyr, Isar, Tubin) of Khongorai and their tributary *kyshtym* clans (Kachin, Sagai, Beltir, Kyzyl, Shor, Kamasin) numbered ~12,8k, while Tuvans and Todzhins numbered ~11,2k and Sayan Samoyeds (Mator, Karagas, Koibal) ~1,2k, giving a native population

of 25,2k (0,0663 persons/km^2) [9][30]. After the Qing Empire moved a large part of the Yenisei Kyrgyz elite to the Ili valley in 1703 and Russia built the Abakan (1707) and Sayansk (1718) forts, the Minusinsk, Achinsk and Kansk forest-steppes became the main grain-growing belt of Yeniseysk Governorate (founded 1822) [37][51]. Under the Kachinskaya, Sagayskaya and Kyzylskaya Steppe Dumas, the native Khakas ("Yenisei Turks") grew from ~14k in the mid-18th century to 37,7k in the 1897 Census, while Russian peasant immigration pushed the total population of Minusinsk Okrug alone to 182,7k (with another 111,5k in Achinsk Okrug and 96,2k in Kansk Okrug) [11][37]. South of the Sayan Range in Qing-ruled Uriankhai (Tuva, *Tangnu-Uryankhai*), banners (*khoshuns*) of Tuvan herders numbered ~50k-55k by the late 19th century, joined from the 1860s onward by Russian Old Believer farmers, gold miners and merchants (~2,5k Russians by 1900AD) [52][53]. Across the entire Sayan, Minusinsk and Tuva region, total population rose from 31,5k in 1700AD to 58k in 1750AD, 105k in 1800AD, 215k in 1850AD and 440k in 1900AD.

Middle and Lower Yenisei

Covering ~2,25M km^2 of northern forest and Arctic tundra across the Middle and Lower Yenisei, Lower Angara, Stony and Lower Tunguska and Taimyr Peninsula, the Middle Siberian basin was inhabited from the Early Holocene (Ust-Kova on the Lower Angara; Tagenar VI on the Taimyr Peninsula, ~6000 cal BC) by river fishers and wild reindeer hunters (rising from ~1,2k in 10000BC to ~4,8k in 4500BC and ~8,5k in 2000BC). During the 2nd millennium BC, Ymyyakhtakh and Pyasina groups worked the world's northernmost bronze foundries at Abylaakh-1 on the Kheta River (71°N, ~1200 cal BC), casting socketed axes from local Norilsk-Talnakh native copper. Iron Age Tsepan and Ust-Chernaya bloomery iron-smelters along the Angara and Stony Tunguska preceded the 17th-century native groups reconstructed by B. O. Dolgikh [9] and N. A. Tuchkova [54]: 10,16k Western Evenks across the Tunguska basins, 5,54k Yeniseian peoples (Inbatsk and Sym Kets, Kott, Arin, Assan, Pumpokol, Yugh) and 4,75k Yenisei-Taimyr Samoyeds (Enets, Nganasan/Tavgi and Yenisei Nenets), totalling 20,45k native inhabitants in 1650AD (0,0091 persons/km^2), or 29,5k in 1700AD including Russian settlers around Yeniseysk (1619), Krasnoyarsk (1628) and Turukhansk.

In the 18th and 19th centuries, farming villages spread along the Yeniseysk-Krasnoyarsk-Nizhneudinsk-Bratsk river and post-road routes, while the discovery of rich alluvial gold in the North and South Yenisei taiga (the Yeniseysk gold rush of the 1830s-1850s) drew tens of thousands of seasonal miners [11][51]. In the northern Turukhansk Krai (10,9k inhabitants in the 1897 Census), native Evenk, Ket, Selkup, Enets, Nganasan and Dolgan populations remained steady at ~10k-12k, while southern Kott, Arin and Assan groups were absorbed into Russian and Khakas communities [12]. Overall population in the Middle and Lower Yenisei basin rose from 48k in 1750AD to 78k in 1800AD, 132k in 1850AD and 215k in 1900AD.

Vitim and Aldan Basins

Multi-layered river terrace sites along the Aldan, Olekma and Vitim rivers (Belkachi I with 23 cultural layers, Ust-Timpton with 26 layers, Sumnagin I with 38 layers and Ust-Mil II) provide the main radiocarbon sequence for interior North-East Asia [55]. During the late Ice Age Dyuktai (~14000-8500 cal BC) and Early Holocene Sumnagin Mesolithic (*Sumnaginskaya kultura*, ~8500-5000 cal BC, 46 ^{14}C dates), microblade hunters living in light surface *chum* tents (3-4 m stone-ring hearths) hunted elk (*Alces alces*) and wild reindeer along the Aldan and Vitim valleys, with regional population growing from ~1k in 10000BC to ~1,8k in 8000BC and ~2,8k in 6000BC. During the subsequent Syalakh (~5000-3800 cal BC, 34 dates), Belkachi (~3800-2200 cal BC, 42 dates) and Ymyyakhtakh (~2200-1000 cal BC, 90 dates) cultures, net-impressed, cord-marked and waffle-stamped pottery accompanied semi-subterranean winter houses and composite bone/antler bows (Belkachi I, Tuoi-Khaya), raising the regional population from ~4k in 4500BC to ~6,5k in 2000BC [55][56].

During the Bronze Age Ust-Mil culture (~1300-400 cal BC, 48 dates) and the Early Iron Age, local bronze casting (Ust-Mil II) and ironworking supported ~7,4k people in 1200BC and ~9,6k in 1200AD. In 1650AD [9], the Vitim and Aldan basins held ~8,8k Evenks and ~1,5k southern Evens (10,3k total; 0,0092 persons/km^2). In the 18th and early 19th centuries, Sakha cattle-and-horse herders settled along the Amga and lower Aldan valleys, raising the regional population to 13,5k in 1750AD and 18k in 1800AD. From the 1840s to 1890s, the opening of the Olekma and Vitim (Bodaibo) placer goldfields drew thousands of Russian, Sakha and Buryat mineworkers and carters, lifting the regional total from 26k in 1850AD to 48k in 1900AD [11][36].

Central Yakutia and Middle Lena

Throughout the Neolithic, Bronze Age and Early Iron Age (Syalakh, Belkachi, Ymyyakhtakh, Ust-Mil, Kullaty, Pokrovsk), the Middle Lena, Lower Lena (Siktyakh, Zhigansk) and Vilyuy river basins were occupied by forest hunting and fishing bands whose population grew gradually from ~1,1k in 10000BC to ~4,6k in 4500BC, ~7,5k in 2000BC and ~9,6k in 1AD [55].

Between the 11th and 14th centuries AD, Turkic-speaking Sakha (*Yakuty*) ancestors migrating from the Baikal-Angara region brought hardy northern horse and cattle breeding into the Central Yakutian Lowland (the Tuymaada, Erkeeni and Ensieli valleys along the Middle Lena and the Lena-Amga interfluve between 61°N and 63,5°N), making use of the lake-meadow depressions (*alaas*) scattered through the permafrost forest [39][57][58]. During the Kulun-Atakh culture (*Kulun-Atakhskaya kultura*, 14th-16th centuries AD), excavations at Kulun-Atakh, Ugar, Sayylyk and the 15th-century site of Ayyy-Taybyt in Yakutsk uncovered timber winter houses (*balagan*) with clay-plastered hearths (*komulok*), birch-bark summer tents (*urasa*), bloomery iron furnaces and hay-storage corrals [39]; contemporary “Small Houses” (*Kultura malykh domov*) sites around the edges of the *alaas* lowland represent neighbouring Evenk hunter-fishers adopting Sakha tools and speech [39].

When Russian Cossacks founded Yakutsk Ostrog in 1632, over 80% of the Sakha lived within a 25k km^2 *alaas* core between the Middle Lena, lower Aldan and lower Amga (local density > 1,1 persons/km^2). From the *yasak* books of 1634-1690, B. O. Dolgikh [9] and V. V. Ushnitskiy [10][31] reconstructed 28,47k Sakha across 35 *volosts* (Kangalasskaya ~4,8k, Meginskaya ~3,6k, Borogonskaya ~3,3k, Baturusskaya ~3,1k, Namskaya ~2,5k, Betunskaya ~1,6k and Bayagantayskaya, plus Vilyuy and Olekma settlements established from the 1630s onward) alongside ~5,7k Vilyuy and Lower Lena Evenks, giving a native population of 34,17k in 1650AD (37,5k in 1700AD including Russian servicemen and Lena peasant farmers). Despite periodic smallpox outbreaks in the 18th century, the Sakha expanded steadily across the entire Vilyuy basin (Suntar, Nyurba, Verkhnevilyuysk) and the Olekma valley, draining lakes (*khoruu*) to create new hay meadows and adopting barley and rye growing from Russian peasants settled along the Lena post stations [10][36]. Poll-tax revisions and Yakut Steppe Duma records show the Sakha growing from ~28,5k in 1650 to ~80k at the 3rd Revision (1763), ~125k at the 7th Revision (1816), ~185k at the 10th Revision (1858) and 225,8k across Yakutsk Oblast in the 1897 Census (83,7% of the oblast’s 269,9k residents, with Russians accounting for ~30k and Evenks/Evens/Yukaghirs ~14k) [12][36][59]. Within Central Yakutia and the Middle Lena (Yakutsk, Vilyuysk and Olekminsk *okrugs*), total population increased from 62k in 1750AD to 112k in 1800AD, 175k in 1850AD and 225k in 1900AD.

Yana, Indigirka and Kolyma Basins

At the High Arctic Zhokhov site (76°06’N, De Long Archipelago, ~7000-5800 cal BC), excavations by V. V. Pitulko (over 540 m^2 of permafrost trench and more than 70 ^{14}C dates) uncovered 13 semi-subterranean driftwood and whalebone dwellings at 76°N, along with remains of more than 150 polar bears (*Ursus maritimus*) and 200 reindeer, wooden dog sleds and obsidian blades carried over 1,5k km from Lake

Krasnoye on the lower Anadyr, representing a resident hunting band of ~25-50 people [56]. Along the Upper and Lower Kolyma and Indigirka, S. B. Slobodin [60][61] studied Mesolithic and Neolithic campsites at Burkhala, Rodinka and Labuya (Burkhala yielded over 6k stone artefacts around stone-lined hearths; the Rodinka II burial contained 36 bone/antler arrowheads and 130 adzes and knives), matching a gradual rise in population from ~0,7k in 10000BC to ~3,8k in 2000BC and ~4,9k in 1AD.

For 1650AD, B. O. Dolgikh [9] reconstructed 3,29k Yukaghirs within the Yana, Indigirka and Kolyma basins (~775 Yana Yukaghirs, ~1,2k Indigirka and Alazeya Yukaghirs and ~1,32k Kolyma and Omolon Yukaghirs, out of 4,78k total Yukaghirs including 1,49k Anadyr Chuvans, Khodyntsy and Anauls), ~2,4k Verkhoyansk-Kolyma Evens (*Lamuty*) and ~0,3k northern Sakha, giving 5,99k native inhabitants (0,0039 persons/km^2). Over the 18th and 19th centuries, smallpox epidemics and assimilation reduced the registered Yukaghirs to fewer than 1k by 1897, while northern Sakha cattle-and-horse herders and Even reindeer herders expanded across the Yana, Indigirka and Kolyma valleys [12][36]. In the 1897 Census, Verkhoyansk Okrug recorded 14,3k inhabitants and Kolymsk Okrug 7,9k inhabitants, bringing the regional population from 6,3k in 1700AD to 8k in 1750AD, 11,5k in 1800AD, 16,5k in 1850AD and 22,2k in 1900AD [11][59].

Amur, Ussuri and Primorye

Early Pottery, Pit-House Villages and Millet Farming

The Lower and Middle Amur preserve some of the oldest pottery vessels in the world within the Osipovka (Gasya, Khummi, Goncharka-1) and Gromatukha cultures (13,3k-12,3k uncal BP / ~14200-12200 cal BC), used by river fishers catching seasonal runs of chum and pink salmon [22][23][62]. During the Mariinskoe, Malyshevo, Kondon and Voznesenovka phases (~6000-1500 cal BC), large permanent fishing villages lined the Lower Amur [3]; on Suchu Island (3,7 ha), more than 142 circular and oval semi-subterranean house depressions (over 25 excavated, 35-150 m^2 floor area) housed ~120-250 residents at a time. Along Peter the Great Gulf in southern Primorye, the Boisman culture (Boisman-1 and 2, ~4800-3000 cal BC) left oyster shell middens with ring cemeteries (over 40 skeletons at Boisman-2, $\delta^{15}\mathrm{N} = +16{,}3‰$) [24][63]. During the Zaisanovka phase (~3000-1300 cal BC), direct AMS dates on charred foxtail and broomcorn millet (*Setaria italica*, *Panicum miliaceum*) at Krounovka-1, Novoselishche-4 and Bogolyubovka-1 (over 25 excavated pit-houses) date the arrival of dryland farming in Primorye to ~2900-2500 cal BC, supporting a rise in regional population from ~17,5k in 4500BC to ~25,5k in 3000BC and ~34k in 2000BC [22][64].

Iron Age Villages, the Bohai Kingdom and Jurchen Fortresses

During the Early Iron Age (~1000BC-400AD) and Mohe period (4th-10th centuries AD), coastal Yankovsky shell-midden farming villages (Peschany-1 with over 30 houses), inland Krounovka settlements with heated *kan* flues and Middle Amur Urily and Poltse villages (Poltse I contained 10 large burned houses, each holding 50-100 ceramic storage jars) raised the population of the Amur-Primorye basin to ~55k in 500BC, ~66k in 1AD and ~72k in 500AD (the Troitskoye Mohe cemetery on the Middle Amur yielded over 218 graves). Under the Bohai Kingdom (*Gosudarstvo Bokhay*, 698-926AD), more than 20 walled Bohai towns and over 200 rural villages were built in Primorye [25][65], including the stone-walled port city of Kraskinskoye (~13 ha, 1,38k m perimeter wall, Buddhist temple, roof-tile kilns and residential wards housing ~1,5k-3k inhabitants), Koksharovka-1 (16 ha) and Maryanovka (15 ha), supporting ~86k people across the region around 800AD.

Under the Jurchen Jin Empire (1115-1234AD) and the Eastern Xia State (1215-1233AD), over 30 Jurchen mountain fortresses (*gorodishcha*) and valley towns were built across Primorye [8][26]. At Krasnoyarovskoye Gorodishche near Ussuriysk (~180 ha enclosed by an 8 km stone-and-earth rampart), Puxian Wannu's northern capital of Kaiyuan contained a walled inner palace compound and terraced

residential quarters with a wartime urban population of ~10k-20k [8]. At Shayginskoye Gorodishche (47,6 ha), excavations uncovered more than 310 dwellings heated by *kan* flues, 14 ironworking and bronze workshops and an inner citadel, supporting ~3k-4,5k permanent inhabitants (~550 total houses) [26]. Ananyevskoye (31 ha), Nikolaevskoye (27 ha), Yekaterinovskoye (27 ha), Skalistoye (22 ha) and Lazovskoye (19 ha) each held 1,5k-3,5k residents, bringing the Amur-Primorye basin to a medieval peak of ~95k around 1200AD, before the Mongol conquest of 1233AD and later Ming and early Qing resettlements reduced the population to ~54k by 1500AD.

Qing Deportations, Nerchinsk Mining and Far Eastern Settlement

When Vasily Poyarkov (1643-1646) and Yerofey Khabarov (1649-1653) descended the Amur, they found fortified timber towns (Albazin, Guigudar with ~1k residents, Tolga, Atchansky) inhabited by plough-farming Daurs (~9,7k along the Upper/Middle Amur and Zeya) and Duchers (~14k along the Middle Amur), alongside ~9k Amur Evenks and Negidals, ~7,5k Nanai (5,2k) and Ulchi (2,3k), ~4,3k mainland Nivkhs and ~2,4k Udege and Orochi, giving a native total of 46,9k in 1650AD (0,0479 persons/km^2) [9][32][33]. Between 1654 and 1658, the Qing government forcibly moved ~20k-24k Daurs and Duchers southward to the Nonni and Sungari valleys in Manchuria to deprive Russian Cossacks of local grain supplies, reducing the population of the Amur-Primorye region to 28,5k by 1700AD [13][33].

Between 1689 (Treaty of Nerchinsk) and 1858, Russian territory within the Amur drainage was limited to the upper Shilka and Argun valleys of eastern Transbaikalia (the Nerchinsk Mining District, founded 1704), where silver-lead mines, smelting works and peasant villages raised the population of the Amur basin to 62k in 1750AD, 102k in 1800AD and 155k in 1850AD (while the Middle/Lower Amur and Ussuri basins held only ~17,7k native Nanai, Ulchi, Nivkh, Evenk, Udege and Manchu-Chinese residents in 1858) [13]. Following the 1858 Treaty of Aigun and 1860 Treaty of Beijing, peasant and Cossack settlement transformed Priamurye and Primorye [13][66][67]. The Russian government established the Amur and Ussuri Cossack Hosts along the border rivers and, after starting regular steamship voyages of the Volunteer Fleet (*Dobroflot*) from Odessa to Vladivostok in 1883 and beginning construction of the Ussuri and Trans-Siberian Railways in 1891, settled over 150k Ukrainian and Russian peasants in the Zeya-Bureya plain and Khanka-Ussuri lowland [13][67]. Fast-growing cities included Vladivostok (28,9k in 1897), Blagoveshchensk (32,8k), Khabarovsk (15k) and Nikolsk-Ussuriysky (10,9k), alongside Korean farming villages (~26k Koreans in 1897) and Chinese merchants and gold miners (~57k Chinese in 1897) [11][66]. By 1897-1900AD, Amur Oblast counted 120,3k inhabitants in 1897 (130,4k in 1900), the southern districts of Primorskaya Oblast (South Ussuri, Khabarovsk and Udsk/Nikolaevsk *okrugs*) counted ~207k in 1897 (~245k in 1900) and the Nerchinsk and Shilka-Argun *okrugs* of eastern Transbaikalia counted ~215k, giving the Amur, Ussuri and Primorye region a total population of 590k in 1900AD [11][13].

Kamchatka, Sakhalin and the Kuril Islands

At Ushki I and V in the central Kamchatka River valley, Goebel, Waters and Dikova [4] dated Layer VII (over 10 double-chambered surface houses, stemmed points and a dog burial) to ~11000-10700 cal BC and Layer VI (over 40 semi-subterranean pit-houses with stone-lined entrance tunnels and microblades) to ~10300-9800 cal BC. During the Late Holocene (Tarya and Old Itelmen phases), more than 120 fortified river villages (*ostrozhki*) lined the Kamchatka, Bolshaya, Avacha and Tigil rivers, each housing 50-100+ people in winter *zimniki* and raised summer *balagans*. For 1650AD, Dolgikh [9], Zuev [68] and Tolkacheva [34] reconstructed 12,68k Itelmens (*Kamchadaly*), ~3,5k northern Kamchatka and Karaga Koryaks and ~2,9k Okhotsk-coast Evens.

On Sakhalin Island (160 ^{14}C dates from 74 sites), Kuzmin, Vasilevski et al. [69] traced continuous occupation from Early Neolithic blade sites (~7000 cal BC) and Middle/Late Neolithic Imchin pit-house

villages to the Susuya (~500BC-400AD) and Okhotsk (~400-1200AD) coastal sea-mammal hunting cultures (~2k Sakhalin Nivkhs, ~2,8k Sakhalin Ainu and ~0,4k Uilta/Evenks in 1650AD). Across the Kuril Islands (380 radiocarbon dates), Fitzhugh et al. [27] identified two major coastal settlement waves (Epi-Jomon at ~400BC-400AD and Okhotsk Culture at ~650-1200AD, followed by Kuril Ainu communities of ~1,4k residents; Figure 2, Panel B). Across Kamchatka, Sakhalin and the Kuril Islands, population rose from ~1,8k in 10000BC to ~13,5k in 2000BC, ~20k in 1AD, ~24k in 800AD and 25,68k in 1650AD (0,0395 persons/km^2).

Following Russian conquest in 1697-1706, heavy transport requisitions, armed uprisings (1706, 1731, 1746) and the smallpox epidemic of 1768-1769 (which killed roughly two-thirds of the Itelmens) reduced the regional population from 26,2k in 1700AD to 18,5k in 1750AD and 16k in 1800AD [34][68]. In the second half of the 19th century, after the 1875 Treaty of Saint Petersburg recognised Russian rule over all of Sakhalin in exchange for ceding the Kuril Islands to Japan, the imperial government expanded the Sakhalin penal colony (*katorga*). In the 1897 Census, Sakhalin Island recorded 28,1k inhabitants (20,4k men and 7,7k women, including ~4,3k native Nivkhs, Ainu and Uilta alongside Russian convicts, exiles and guards), while Petropavlovsk (Kamchatka) Okrug counted 8,4k, Okhotsk Okrug 4,7k, the Commander Islands 0,65k and the Kuril Islands ~4,1k, bringing the region to 46k in 1900AD [11][13].

Chukotka, Anadyr and the Bering Strait

While interior Chukotka and the Anadyr basin were occupied by wild-reindeer hunters at river crossings (Ust-Belaya, Chirovoe, ~3000-500 cal BC), the Bering Strait headlands (Uelen, Ekven, Sireniki, Chini, Enmynytnyn) supported dense coastal sea-mammal and bowhead whaling villages of the Old Bering Sea (~200BC-800AD), Birnirk (~500-1000AD), Punuk (~700-1400AD) and Thule (~1000-1700AD) cultures [28][29][70]. Excavations at Ekven (over 330 graves and more than 350 individuals) and Uelen (over 200 graves) and reconstructions of semi-subterranean bowhead-whalebone and driftwood winter houses (*nynglu*, holding 15-30 occupants each) indicate coastal villages of 150-400 residents [28][29]. Between ~1300 and 1650AD, interior Chukchi (*Chauchu*) and Chavchuven Koryaks developed large-herd tundra reindeer herding. In 1650AD [9][32], the Chukotka, Anadyr and Bering Strait region held 7,25k mainland Koryaks (settled coastal *Nymylan* and nomadic *Chavchuven*), 4k Siberian Yupik, 2,5k Chukchi and 1,49k Anadyr Yukaghirs (Chuvans, Khodyntsy, Anauls), totalling 15,24k native inhabitants (0,0133 persons/km^2).

During the 18th and 19th centuries, following the withdrawal of the Anadyr garrison in 1764 and the opening of peaceful trade at the Anyuy Fair (1788), tundra reindeer-herding Chukchi expanded their pastures westward toward the Kolyma and southward into Koryak lands, growing from ~2,5k in the mid-17th century to ~11,8k by 1897 [38]. In the 1897 Census (carried out in Anadyr and Chukotka with the help of exiled ethnographer V. G. Bogoraz), Anadyr Okrug (12,1k) and the northern mainland part of Gizhiga Okrug (7,5k total) recorded ~18k inhabitants across the Chukotka-Anadyr region (~11,8k Chukchi, ~3,8k mainland Koryaks, ~1,3k Siberian Yupik, ~0,6k Chuvans and Yukaghirs and ~0,5k Russians), reflecting gradual growth from 15,5k in 1700AD to 15,8k in 1750AD, 16,5k in 1800AD, 17,2k in 1850AD and 18k in 1900AD [12][38].

Population Estimates

Combining the archaeological site benchmarks (Table 1) with 17th-century *yasak* rolls, 18th- and 19th-century Poll-Tax Revisions and the 1897 Census yields the long-run population trajectories across the Yenisei-Baikal, Lena-Kolyma and Pacific basins shown in Figure 3 and Tables 3 and 4.

At the time of initial Russian fur-tribute (*yasak*) assessment around 1650AD, Eastern and Pacific Siberia supported 220,53k native inhabitants across the nine drainage regions (Figure 4 and Table 2), with the

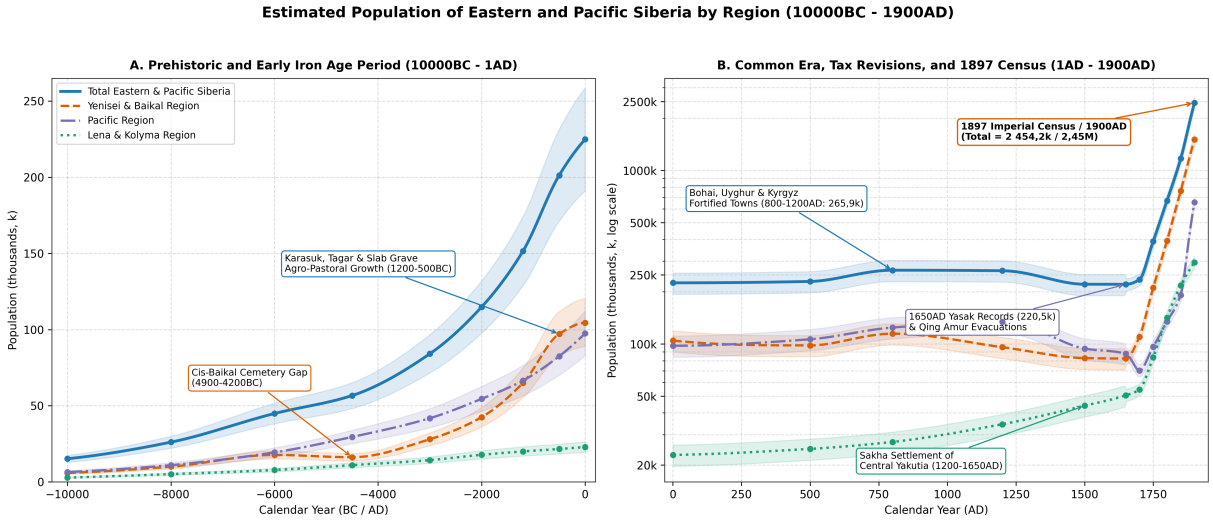


Figure 3. Panel A traces the long-run population of the Yenisei-Baikal, Lena-Kolyma and Pacific basins from early Holocene fishing camps through the Tagar-Yankovsky Iron Age (201,3k in 500BC) and medieval Kyrgyz-Uyghur and Bohai-Jurchen peaks (265,9k in 800AD and 264,1k in 1200AD). Panel B shows the expansion from 234,8k in 1700AD to 2,45M in 1900AD driven by Buryat and Sakha pastoral growth and Russian peasant settlement along the Moscow Highway and in the Amur-Ussuri valleys.

largest concentrations in the Amur-Primorye basin (46,9k), the Baikal-Selenga steppes (36,6k) and the Central Yakutian *alaas* lowland (34,17k).

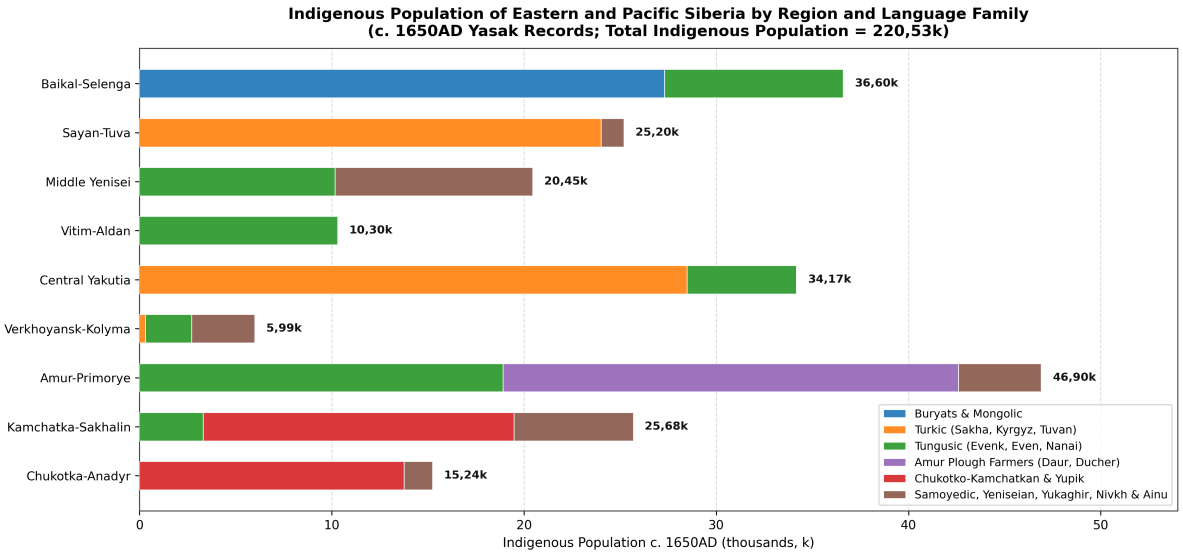


Figure 4. Panel A breaks down the 220,53k native inhabitants reconstructed from mid-17th-century yasak registers across the nine drainage regions, led by the Amur-Primorye (46,9k), Baikal-Selenga (36,6k) and Central Yakutian Middle Lena (34,17k) basins. Panel B groups the 1650AD native population by language family and primary subsistence (Sakha, Buryat, Kyrgyz and Tuvan pastoralists; Daur and Ducher farmers; Evenk, Even and Yukaghir taiga hunter-herders; and Itelmen, Koryak, Nivkh, Ainu and Yupik coastal fishers).

Site / Complex	Reg.	Culture & Calibrated Date	Area & Excavated Features	Site Population
Shamanka II (SW Baikal)	Baikal	Early Neolithic Kitoi (5650-4950BC)	112 graves, MNI = 155 (116 AMS dates)	35-60 band
Lokomotiv (Angara)	Baikal	Early Neolithic Kitoi (5750-5000BC)	91 graves, MNI = 149 (FRE-corrected)	50-80 band
Khuzhir-Nuge XIV (Baikal)	Baikal	Early Bronze Glazkovo (2750-1950BC)	79 slab graves, MNI = 89 (82 AMS dates)	30-50 band
Ivolga Gorodishche	Baikal	Xiongnu Frontier (150BC-50AD)	7,2 ha (4 ramparts); 54 kan houses, 216 graves	2,5k-3k town
Torgazhak (Minusinsk)	Sayan-Tuva	Late Bronze Karasuk (1100-900BC)	1,5 ha village; 6 timber houses (100-200 m^2)	120-180 village
Arzhan-1 & Tunnug-1	Sayan-Tuva	Early Scythian Aldy-Bel (9th-8th c. BC)	110-120 m royal kurgans; 70 chambers, 160+ horses	1k-1,5k builders
Oglakhty Cemetery	Sayan-Tuva	Tashtyk Culture (1st-7th c. AD)	Permafrost graves + collective crypts (20-100+ MNI)	150-300 clan
Por-Bajin & 17 Forts	Sayan-Tuva	Uyghur Khaganate (Summer 777AD)	3,5 ha palace (Por-Bajin); 22 ha (Shagonar III)	8k-15k line
Ayyy-Taybyt & Sayylyk	Mid. Lena	Sakha Kulun-Atakh (14th-16th c. AD)	Alaas timber balagan dwellings & cattle corrals	15-40 homestead
Zhokhov Site (76 N)	Arctic	Arctic Mesolithic (7000-5800BC)	13 driftwood/ whalebone houses (>540 m^2 trench)	25-50 band
Suchu Island (Amur)	Amur	Neolithic Villages (6000-1500BC)	3,7 ha island; 142 pit-houses (25+ excavated)	120-250 village
Boisman-2 (Primorye)	Amur	Middle Neolithic (4400-3300BC)	Shell midden; 5 houses + 28 graves (MNI > 40)	35-60 band
Kraskinskoye (Posyet)	Amur	Bohai Port City (8th-10th c. AD)	13 ha walled city (1,38k m perimeter), temple	1,5k-3k city
Krasnoyarovskoye	Amur	Jurchen Capital Kaiyuan (1215-1233AD)	180 ha mountain capital (8 km wall), Forbidden City	10k-20k city
Shayginskoye	Amur	Jurchen Fortress Town (1215-1233AD)	47,6 ha fortress; >310 kan houses, 14 workshops	3k-4,5k town
Ushki Lake I & V	Kamchatka	Terminal Pleistocene (11000-9800BC)	Layers VII-VI: >50 surface & pit dwellings	30-60 band
Ekven & Uelen	Bering	Old Bering Sea / Punuk (200BC-1400AD)	>530 graves (Ekven >330, Uelen >200) + nynglu houses	150-400 village

Table 1. Calibrated dates, walled or site areas, excavated house and burial counts and estimated community sizes for 17 reference archaeological complexes across Eastern and Pacific Siberia.

Region	Area (km²)	1650 Native	1900 Total	Main Peoples and Districts (1650AD and 1897/1900AD)
Baikal-Selenginsky	650k	36,6k	850k	1650: Buryats 27,3k, Evenks 9,3k. 1897/1900: Buryats 288,6k, Russians & Semeiskie 548k, Evenks 13k
Sayan-Tuvinsky	380k	25,2k	440k	1650: Kyrgyz/Khakas 12,8k, Tuvans 11,2k. 1897/1900: Minusinsk-Achinsk-Kansk Russians 345k, Khakas 37,7k, Tuva 55k
Middle Siberian	2,25M	20,45k	215k	1650: Evenks 10,16k, Kets/Yeniseians 5,54k, Samoyeds 4,75k. 1897/1900: Yeniseysk/Angara Russians 192k, Indig. 23k
Subtotal Yenisei & Baikal	3,28M	82,25k	1,5M	Yenisei & Baikal Region (82,25k native in 1650AD; 1,5M total by 1900AD)
Vitimsko-Aldansky	1,12M	10,3k	48k	1650: Evenks 8,8k, Evens 1,5k. 1897/1900: Vitim-Olekma goldfields 22k, Amga-Aldan Sakha 14k, Evenks/Evens 12k
Prilensky	1,38M	34,17k	225k	1650: Sakha 28,47k (35 volosts), Evenks 5,7k. 1897/1900: Sakha 198k in Middle Lena (225,8k in oblast), Russians 20k
Verkhoyansk-Kolymsky	1,55M	5,99k	22,2k	1650: Yukaghirs 3,29k, Evens 2,4k, Sakha 0,3k. 1897/1900: Verkhoyansk & Kolymsk okrugs (Sakha, Evens, Yukaghirs, Russians)
Subtotal Lena & Kolyma	4,05M	50,46k	295,2k	Lena & Kolyma Region (50,46k native in 1650AD; 295,2k total by 1900AD)
Amursky	980k	46,9k	590k	1650: Duchers 14k, Daurs 9,7k, Evenks/Nanai/Nivkh 23,2k. 1900: Amur (130k), S. Primorye (245k), Nerchinsk (215k)
Kamchatka-Sakhalinsky	650k	25,68k	46k	1650: Itelmens 12,68k, Koryaks 3,5k, Ainu 4,2k, Evens/Nivkh 5,3k. 1897/1900: Sakhalin (28,1k), Kamchatka-Okhotsk-Kurils (17,9k)
North Pacific	1,15M	15,24k	18k	1650: Koryaks 7,25k, Yupik 4k, Chukchi 2,5k, Chuvans 1,49k. 1897/1900: Chukchi 11,8k, Koryaks 3,8k, Yupik 1,3k
Subtotal Pacific Region	2,78M	87,82k	654k	Pacific Region (87,82k native in 1650AD; 654k total by 1900AD)

TOTAL	10,11M	220,53k	2,45M	Full Eastern & Pacific Siberia (220,53k native in 1650AD; 2,45M total in 1900AD)

Table 2. Drainage areas, 1650AD native populations reconstructed from Russian yasak books and 1897/1900AD census totals across the nine regions of Eastern and Pacific Siberia, with ethnic and district breakdowns.

Year	Baikal	Sayan	Yenisei	Vitim	Lena	Kolyma	Amur	Kamch.	Chukot.	Total
10000BC	2,2k	2,5k	1,2k	1k	1,1k	0,7k	3,5k	1,8k	1,2k	15,2k
8000BC	3,8k	4k	2,2k	1,8k	2k	1,3k	6k	3k	2k	26,1k
6000BC	8,2k	6k	3,5k	2,8k	3,2k	1,8k	11k	5,2k	3,2k	44,9k
4500BC	3,2k	8,2k	4,8k	4k	4,6k	2,4k	17,5k	7,5k	4,5k	56,7k
3000BC	7,5k	14k	6,5k	5,2k	6k	3,1k	25,5k	10,5k	5,8k	84,1k
2000BC	12k	22k	8,5k	6,5k	7,5k	3,8k	34k	13,5k	7k	114,8k
1200BC	18,5k	36k	10,5k	7,4k	8,4k	4,2k	43k	15,5k	8k	151,5k
500BC	27k	58k	12,2k	8k	9k	4,6k	55k	18k	9,5k	201,3k
1AD	36k	55k	13,5k	8,4k	9,6k	4,9k	66k	20k	11,5k	224,9k
500AD	31k	52k	15k	8,8k	10,8k	5,2k	72k	21,5k	12,8k	229,1k
800AD	34k	64k	16,5k	9,2k	12,5k	5,5k	86k	24k	14,2k	265,9k
1200AD	35,5k	42k	18,2k	9,6k	19k	5,8k	95k	24,5k	14,5k	264,1k
1500AD	35k	28k	19,8k	10k	28k	6k	54k	25k	14,9k	220,7k
1650AD (Indig.)	36,6k	25,2k	20,5k	10,3k	34,2k	6k	46,9k	25,7k	15,2k	220,53k
1700AD	49k	31,5k	29,5k	10,8k	37,5k	6,3k	28,5k	26,2k	15,5k	234,8k
1750AD	105k	58k	48k	13,5k	62k	8k	62k	18,5k	15,8k	390,8k
1800AD	210k	105k	78k	18k	112k	11,5k	102k	16k	16,5k	669k
1850AD	415k	215k	132k	26k	175k	16,5k	155k	19k	17,2k	1,17M
1900AD	850k	440k	215k	48k	225k	22,2k	590k	46k	18k	2,45M

Table 3. Population of the nine drainage basins of Eastern and Pacific Siberia at 19 benchmark dates from the end of the Ice Age to 1900AD, with combined regional totals.

Year	Baikal	Sayan	Yenisei	Vitim	Lena	Kolyma	Amur	Kamch.	Chukot.	Overall
10000BC	0,0034	0,0066	0,0005	0,0009	0,0008	0,0005	0,0036	0,0028	0,001	0,0015
6000BC	0,0126	0,0158	0,0016	0,0025	0,0023	0,0012	0,0112	0,008	0,0028	0,0044
4500BC	0,0049	0,0216	0,0021	0,0036	0,0033	0,0015	0,0179	0,0115	0,0039	0,0056
2000BC	0,0185	0,0579	0,0038	0,0058	0,0054	0,0025	0,0347	0,0208	0,0061	0,0114
500BC	0,0415	0,1526	0,0054	0,0071	0,0065	0,003	0,0561	0,0277	0,0083	0,0199
1AD	0,0554	0,1447	0,006	0,0075	0,007	0,0032	0,0673	0,0308	0,01	0,0222
800AD	0,0523	0,1684	0,0073	0,0082	0,0091	0,0035	0,0878	0,0369	0,0123	0,0263
1200AD	0,0546	0,1105	0,0081	0,0086	0,0138	0,0037	0,0969	0,0377	0,0126	0,0261
1650AD	0,0563	0,0663	0,0091	0,0092	0,0248	0,0039	0,0479	0,0395	0,0133	0,0218

1750AD	0,1615	0,1526	0,0213	0,0121	0,0449	0,0052	0,0633	0,0285	0,0137	0,0387
1800AD	0,3231	0,2763	0,0347	0,0161	0,0812	0,0074	0,1041	0,0246	0,0143	0,0662
1850AD	0,6385	0,5658	0,0587	0,0232	0,1268	0,0106	0,1582	0,0292	0,015	0,1158
1900AD	1,31	1,16	0,0956	0,0429	0,163	0,0143	0,602	0,0708	0,0157	0,2428

Table 4. Population density (in persons per km²) across the nine drainage basins of Eastern and Pacific Siberia at 13 benchmark dates, contrasting the southern steppe and Amur farming corridors with the northern taiga and Arctic tundra.

Limitations

Four practical constraints affect the uncertainty margins (+/- 15%-25% before 1650AD; +/- 5%-10% between 1700 and 1900AD) around the regional estimates. First, whereas stone-slab burial mounds in Minusinsk and Tuva, shell middens in Primorye (Boisman) and permafrost whalebone villages along the Bering Strait (Ekven, Uelen) preserve durable dwellings and hundreds of burials, acidic soils in the northern larch forest dissolve unburned bone within a few centuries. Forest estimates before the 17th century therefore depend on stratified hearth counts (Aldan and Kolyma river terrace sequences) and comparison with 17th-century *yasak* densities [55][60]. Second, in both Lake Baikal (offsets of 300-700 ^{14}C years) and the Northwest Pacific and Bering Strait (marine reservoir offsets ΔR of 250-500 ^{14}C years), human-bone dates from fish and seal eaters appear several centuries too old unless paired with land-animal bone or corrected using carbon and nitrogen isotopes [2][14].

Third, in Uyghur Tuva (17 fortresses dated 750-840AD) and Jurchen Eastern Xia Primorye (Krasnoyarovskoye at 180 ha and Shayginskoye at 47,6 ha during 1215-1233AD), large walled enclosures sheltered military garrisons and refugees displaced from Manchuria and Mongolia; regional estimates therefore reflect sustainable valley farming and herding populations rather than projecting wartime fortress densities across the entire countryside [7][8]. Fourth, before M. M. Speransky's 1822 administrative reform and the creation of native Steppe Dumas, early Poll-Tax Revisions (1719-1816) often carried forward outdated *yasak* assessments for remote tundra and forest clans and omitted unpacified Chukchi and Qing-ruled Uriankhai Tuvans [13][38][53]. Our 1750 and 1800AD estimates correct for these omissions by combining revision returns with 1763 *Yasak* Commission schedules and comparison with the 1897 Census [12][36].

Conclusion

From the end of the Ice Age (10000BC) to the close of the 19th century (1900AD), the population of Eastern and Pacific Siberia grew through five distinct stages. Early and Middle Holocene fishing and coastal-hunting communities expanded along the Baikal-Angara rapids (Kitoi), the Lower Amur (Osipovka to Suchu Island) and the Pacific coast (Boisman, Ushki), interrupted in Cis-Baikal by the 4900-4200 cal BC cemetery gap. During the Bronze and Early Iron Ages, millet farming, herding and metallurgy in the southern steppes (Karasuk, Tagar, Aldy-Bel, Slab Grave and Xiongnu) and Primorye (Zaisanovka to Yankovsky and Krounovka) raised the combined population past 200k by 500BC. In the Middle Ages, walled towns and fortresses of the Yenisei Kyrgyz and Uyghurs in the Sayan-Tuva basins and the Bohai and Jurchen states in Primorye carried the regional total to ~265,9k in 800AD and ~264,1k in 1200AD. Between the 13th and 17th centuries, Sakha horse-and-cattle herders settled the Central Yakutian *alaas* lowland

(reaching 34,17k in 1650AD) while Chukchi and Koryak herders expanded across the north-eastern tundra, bringing the mid-17th-century *yasak* total to 220,53k native inhabitants. Finally, 18th- and 19th-century growth among the Buryats (288,6k in 1897) and Sakha (225,8k in 1897) combined with Russian peasant, Cossack and mining settlement across the Minusinsk, Irkutsk, Transbaikal, Lena-Vitim and post-1858 Amur-Ussuri basins to lift the total population of Eastern and Pacific Siberia to 2,45M by 1900AD.

Appendix

Regional Breakdown

To provide a finer geographic breakdown within the nine drainage regions of Chernykh et al. [1], Table 5 lists 18 sub-areas and their estimated populations in 1650AD and 1900AD.

Area Name	1650AD Pop.	1900AD Pop.	Archaeological Cultures, 17th-c. Groups & 1897/1900AD Districts
Cis-Baikal & Upper Angara	17,6k	430k	Kitoi, Glazkovo, Kurykans; Irkutsk/ Balagansk/Verkholensk Russians & Western Buryats
Transbaikal Steppe (Selenga-Onon)	19k	420k	Slab Grave, Xiongnu (Ivolga); Verkhneudinsk/Chita Russians, Semeiskie & Khori/Selenga Buryats
Minusinsk, Achinsk & Kansk Steppe	14k	385k	Afanasievo, Karasuk, Tagar, Tashtyk, Kyrgyz; Minusinsk-Achinsk-Kansk peasants & 37,7k Khakas
Tuva Basin (Ulug-Khem & Khemchik)	11,2k	55k	Aldy-Bel (Arzhan-1, Tunnug-1), Kokel, 17 Uyghur forts (Por-Bajin); Tuvan khoshuns & Russian settlers
Middle Yenisei, Angara & Tunguska	15,5k	203k	Ust-Kova, Tsepan; Yeniseysk/Angara peasants, goldfields, Tunguska Evenks & Kets
Lower Yenisei & Taimyr Tundra	4,95k	12k	Pyasina/Abylaakh bronze foundries; Turukhansk Krai Nganasans, Enets, Nenets & Dolgans
Aldan-Amga-Maya River Terraces	6,7k	22k	Sumnagin, Syalakh, Belkachi, Ymyyakhtakh, Ust-Mil; Amga-Aldan Sakha & Evenks
Vitim-Olekma Mountain Taiga	3,6k	26k	Vitim-Olekma Neolithic; Bodaibo/Olekma gold-mining settlements & Vitim Evenks
Central Yakutian Alaas Lowland	28k	148k	Kulun-Atakh (Sayylyk, Ayyy-Taybyt); Yakutsk Okrug Sakha (alaas core) & Lena Russian peasants
Vilyuy & Lower Lena Basin	6,17k	77k	Ymyyakhtakh/Ust-Mil fishers; Vilyuysk & Olekminsk Okrug Sakha pastoralists & Evenks
Yana-Indigirka-Kolyma Valleys	5,89k	22,1k	Burkhala, Rodinka; Verkhoyansk & Kolymsk Okrug Sakha, Evens, Yukaghirs & Russians
New Siberian High Arctic Islands	0,1k	0,1k	Early Holocene Zhokhov polar-bear hunters (76 N); seasonal mammoth-ivory & hunting camps

Upper/Middle/Lower Amur & Zeya	36,6k	345k	Osipovka, Gromatukha, Suchu, Poltse; Nerchinsk district (215k) & Amur Oblast (130k)
Primorye, Ussuri & Khanka Plain	10,3k	245k	Boisman, Zaisanovka, Bohai, Jurchen (Krasnoyarovskoye, Shayga); Vladivostok & Ussuri settlers
Kamchatka Peninsula & Okhotsk Coast	19k	13,8k	Ushki I-V, Old Itelmen (>120 ostrozhki); Petropavlovsk & Okhotsk okrugs (Russians, Kamchadals, Koryaks)
Sakhalin Island & Kuril Chain	6,68k	32,2k	Imchin, Susuya, Okhotsk Culture; Sakhalin penal colony (28,1k in 1897), Nivkhs, Ainu & Kurils
Interior Chukotka, Anadyr & Koryak	8,38k	12,5k	Ust-Belaya; Reindeer Chukchi (Chauchu), Chavchuven Koryaks & Anadyr Chuvans
Bering Strait & Coastal Headlands	6,86k	5,5k	Old Bering Sea, Birnirk, Punuk, Thule (Ekven, Uelen); Maritime Chukchi, Yupik & Nymylan Koryaks

Table 5. Division of the nine main drainage regions into 18 ecological sub-areas, comparing 1650AD native yasak populations with 1900AD totals and listing principal archaeological cultures and administrative districts.

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